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Asymptotic estimated digestibility, a new indicator of black soldier fly (*Hermetia illucens*) conversion efficiency in relation to larval density

Short title

Asymptotic estimated digestibility, a new measure of conversion efficiency

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Abstract

Black soldier fly larvae (BSFL) are highly efficient at converting organic substrates into insect biomass suitable for livestock nutrition. BSFL conversion efficiency depends on the type and profile of their feed, and formulation approaches use macronutrient contents to reach a targeted zootechnical performance. However, digestibility varies between ingredients and BSFL digestibility data is lacking due to methods unsuitable for insects living in their feed. This study defined Estimated Digestibility (ED) as the difference between distributed feed and frass macronutrient weight, divided by macronutrient weight in distributed feed, and explored its variation with larval density. Using the same amount of chicken feed, same feeding time and 0 to 29 larvae/cm², ED of dry matter (DM), starch, nitrogen, ether extract (EE), neutral detergent fibre, acid detergent fibre, acid detergent lignin, ash and energy were evaluated, along with survival rate, feed conversion ratio (FCR) and larval individual weight gain on a fresh and dry basis. ED of DM, starch, nitrogen, EE and energy increased with larval density following an asymptotic trend. ED of ash increased between 0 and 11.4 larvae/cm² but decreased for higher densities. Using a logistic model, asymptotic ED of DM (80.3±1.3%), starch (99.0±2.3%), nitrogen (78.6±1.1%), EE (95.3±1.5%), ash (58.4±1.0%) and energy (80.6±1.2%) were determined. No effect of areal density on ED of fibre fractions was detected, presumably due to an inappropriate dosage method. Survival rate and individual weight gain peaked at densities below 5 larvae/cm², FCR (dry to fresh) was improved with increasing density, while FCR (dry to dry) was optimised at 11.4 larvae/cm². Asymptotic ED is a new indicator corresponding to the maximal fraction of macronutrients potentially digestible by BSFL. It should be used to develop a diet formulation approach based on digestible rather than crude macronutrient contents. Further studies should focus on evaluating asymptotic ED in different ingredients.

Keywords

insect nutrition, formulation, macronutrient, alternative protein, Diptera

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1. Introduction

Black soldier fly larvae (BSFL; *Hermetia illucens* (L.); Diptera: Stratiomyidae) can quickly convert organic side-streams and biowaste into valuable proteins and lipids (Barragan-Fonseca *et al.*, 2017) which can be used in animal nutrition (Bosch and Swanson, 2021; Dörper *et al.*, 2021) and have been increasingly produced and studied by insect stakeholders. Both profitability and sustainability of BSFL production depend on their ability to convert feed provided to the insects into larval biomass. Conversion efficiency varies namely with feed, rearing conditions and genetics (Greenwood *et al.*, 2021). Significant variability of zootechnical performances (development time, prepupal weight, larval protein and lipid content, etc.) due to the type of feed and its macronutrient composition (mostly carbohydrates, proteins, lipids, fibres and ash) has been extensively documented (Gold *et al.*, 2018; Hopkins *et al.*, 2021). Barragan-Fonseca *et al.* used the Nutritional Geometry Framework to comprehensively map the impact of feed protein and carbohydrate content on BSFL performance. This work laid the ground for a conceptual diet formulation approach that would allow targeting the appropriate macronutrient profile to reach the expected zootechnical performance of BSFL. However, BSFL fed diets formulated with the same protein and carbohydrate contents but different ingredients showed bioconversion rates ranging from 14.5% to 31.8% (Gold *et al.*, 2020b). This variability is namely due to the use of crude macronutrient fractions for the formulation instead of the more appropriate digestible macronutrient contents. Indeed the digestibility of a macronutrient can significantly vary from one ingredient to another, as extensively documented in conventional livestock species such as pigs (Stein *et al.*, 2016) or poultry (Zaefarian *et al.*, 2015). Still, data on digestibility by BSFL are lacking.

Apparent Digestibility in conventional livestock species is defined on a dry weight basis as the amount of the considered macronutrient in ingesta subtracted by macronutrient in faeces, divided by macronutrient in ingesta (Barret *et al.*, 2005). Its calculation requires (1) estimating the amount of feed ingested (usually by comparing the weights of distributed and refused feed) and (2) collecting pure faeces (not contaminated with feed, urines, bedding, etc.) to perform macronutrient content analysis. BSFL feed in group on moist substrates that also constitute their living environment, and excreta cannot be separated from non-ingested feed, exuviae abandoned during moulting and possibly dead individuals (Seyedalmoosavi *et al.*, 2022). All these components are mingled together in a powdered material called frass. Therefore, both conditions to calculate Apparent Digestibility are challenging to meet. A possible partial solution is to attach an excreta collection device to the bottom end of a larva. Still, collected amounts of pure excreta are tiny and this method is impractical on many larvae (Gold *et al.*, 2020a). Furthermore, unlike mammals that produce faeces and urines separately, insects' metabolic wastes are released in the hindgut via Malpighian tubules (Weihrauch and O'Donnell, 2021). Hence their excreta consists of a mix of faeces (undigested feed material) and urines (metabolic wastes). Accordingly, (Waldbauer, 1968) highlighted the difference between Approximate Digestibility, which is calculated based on macronutrient weight in ingesta and in excreta, and Apparent Digestibility, which is similar but requires subtracting the weight of macronutrients in urines from total excreta to obtain the weight of macronutrients in faeces and thus be consistent with the definition of Apparent Digestibility in conventional livestock species. This refinement is suitable when insects' excreta can be separated from non-ingested feed, which is not the case in BSFL. Yet, some authors analysed frass to gain insight into BSFL nutrition but imprecisely referred to Approximate Digestibility (Klammsteiner *et al.*, 2021) (Permana and Ramadhani Eka Putra, 2018). This misuse is due to confusion between distributed and ingested feed and an unclear definition of BSFL excreta and frass (Bosch *et al.*,

2020). In the present paper, we consider excreta as a mix of faeces and urines, and we use the official definition of frass given by the European Commission: “the mixture of insect excrements with parts of dead insects and feeding substrate” (European Commission, 2021). Hence, we propose a new definition for a digestibility approximation specific to insects growing in their feed, like BSFL: Estimated Digestibility (ED) is calculated by dividing the difference between distributed feed and frass macronutrient weight by the amount of macronutrient in distributed feed.

Frass analyses are often performed in waste management studies to quantify “waste reduction” of dry matter (DM), also called “overall degradation”. These terms have the same definition as ED of DM but are imprecise and unsuitable in the field of animal nutrition. ED of DM ranges from 9 to 85% depending on the type of feed but also rearing conditions, such as feed moisture content, feeding duration or larval density (Lalander *et al.*, 2019; Rehman *et al.*, 2017). The latter’s effect on ED has only been indirectly investigated, with results suggesting that ED is positively associated with larval density (Brits, 2017; Permana and Ramadhani Eka Putra, 2018). This is supported by the hypothesis that a higher number of BSFL is more likely to consume a given amount of feed entirely. It can be assumed that a high density would ultimately result in reaching a maximal ED, corresponding to the full potential of the feed that BSFL can digest. However, this has never been documented. In contrast to ED of DM, values of ED of starch, protein, lipid, fibre and ash are scarce to non-existent.

Consequently, this study proposes a method to measure the maximal ED of a feed for BSFL. Null to extremely high larval densities were used, and an extensive analysis of remaining macronutrients in frass was performed to challenge the impact of density on the ED of each macronutrient. This approach aims to consider the maximal ED of a macronutrient as the digestibility coefficient of that macronutrient in that ingredient when developing a BSFL diet formulation tool.

2. Materials and methods

Two trials were performed with slight changes between Trial 1 and Trial 2.

2.1. Insects and feed used

BSF starter larvae were obtained from Agronutris (Toulouse, France). Before the trial period, neonate larvae were fed chicken feed (CF) (Maisadour - Sud-Ouest Aliment VT100 Poulet Jaune Démarrage, Haut-Mauco, France; Table 1) rehydrated to $64 \pm 0.5\%$ moisture and were kept at around 28°C and 55% relative humidity (RH) for about seven days. They were sent to the laboratory, where they were successively sieved with 2.4 mm and 1.0 mm diameter sieves to standardise their size and remove most of the frass. The larvae were then gently homogenised by hand and three samples of about 1 g were taken and counted manually to measure the average weight of a starter larva (4.8 and 4.9 mg for Trial 1 and Trial 2, respectively) and to calculate the overall weight of larvae to use for each experimental density.

Table 1: Analytical chemical composition of the chicken feed used to feed black soldier fly larvae

	Starch	Nitrogen	Ether extract	Ash	NDF	ADF	ADL	Gross energy (kJ/g DM)
Content (% DM)	42.1 ± 1.9	3.4 ± 0.2	2.8 ± 0.1	6.1 ± 0.2	14.4 ± 0.9	5.7 ± 0.7	1.3 ± 0.9	15.8 ± 0.5

Mean $\pm$ standard deviation. DM: dry matter; NDF: neutral detergent fibre; ADF: acid detergent fibre; ADL: acid detergent lignin. Values were obtained on five samples for NDF, ADF and ADL, four for starch and nitrogen and two for ether extract and ash.

2.2. Experimental design

Experiments were conducted in 16.5×10.6×7 cm plastic containers (Pro-Jet BG110, Saint-Ouen-l'Aumône, France) individually placed in a light pantyhose and closed with a lid with an opening of 16×10 cm to prevent larvae from escaping while allowing air circulation. The feed used was CF rehydrated to 68±0.5% moisture using hot tap water and mixed thoroughly (Electrolux Professional Bermixer Pro 450 W, Saint-Denis, France; maximum power for 2:30 min) right before distribution. Each container received a unique input of feed for the whole experimental period, consisting of a fresh weight (FW) of 420±1 g of CF, i.e. 134 g of dry weight (DW), and a specified amount of starter larvae, covering an extensive range of larval density (Table 2). Each larval density was replicated twice in Trial 1 and thrice in Trial 2. Investigated densities were slightly changed from Trial 1 to Trial 2.

Table 2: Experimental larval densities

Number of larvae	Areal density (larvae/cm ²)	Mass density (larvae/kg DM)	Feed availability (mg DM/larva)	Trial
0	0	0	-	1 & 2
400	2.29	2,976	336.0	1 & 2
600	3.43	4,464	224.0	1
700	4.00	5,208	192.0	2
800	4.57	5,952	168.0	1
1,000	5.71	7,440	134.4	1 & 2
1,500	8.57	11,161	89.6	1 & 2
2,000	11.43	14,881	67.2	1 & 2
3,000	17.14	22,321	44.8	1 & 2
5,000	28.57	37,202	26.9	1 & 2

DM: dry matter

Containers were placed in a climatic chamber at 28±0.5 °C and 75±2 % RH, with a L12:D12 photoperiod (Trial 1: Xylem Analytics WTW TS 1006-i, Weilheim, Germany; Trial 2: Percival Scientific DR-36VL, Perry, USA). In both trials, larvae were kept in their container and allowed to feed for seven days. Containers were relocated eight times to eliminate any effect related to possible climatic heterogeneity within the chamber. No additional feed input, sampling or mixing was performed during the trial period.

2.3. Sampling and analysis

At the end of the seven-day feeding period, all containers were removed from the chamber and weighed. Larvae were manually separated from the frass. Depending on the stickiness of the frass, they were either sieved or removed with forceps. They were rinsed with tap water in a fine sieve, dried with paper towel and weighed collectively. The FW of frass was calculated as the final weight of the container before removing the larvae, subtracted by the weight of the cleaned larvae and the weight of the empty container.

Frass, including material sticking to the inner side of the container, was manually collected and mixed with a spatula. DM content was determined on a 4 g sub-sample using a thermobalance set at 120°C (Radwag MA 50.R, Radom, Poland). The chemical composition of frass and CF was analysed on oven-dried (Memmert UE500, Schwabach, Germany; 60°C until constant weight) and ground (Retsch ZM 1000, Haan, Germany; 1 mm grid) samples. Starch content was determined by colourimetry (Roche Cobas Mira S, Basel, Switzerland; Randox GLUC-PAP Kit, Crumlin, UK) after enzymatic hydrolysis (Merck, 10115-5G-F, Darmstadt, Germany) according to Faisant *et al.*; nitrogen content was determined using the Dumas method (Elementar EA Vario Micro Cube, Langenselbold, Germany); ether extract (EE) was determined with the Soxhlet method using diethyl ether (Tecator Soxtec System HT6, Höganäs, Sweden); ash content was determined gravimetrically by complete combustion at 550°C for eight hours (Nabertherm furnace, Lilienthal, Germany); neutral detergent fibre (NDF), acid detergent fibre (ADF) and acid detergent lignin (ADL) were determined using the filter bag technique (Ankom 220 Fibre Analyzer, Macedon, NY, USA) according to the Van Soest method, after treatment with acetone, α -amylase and pronase to prevent interactions with excess lipids, starch and proteins respectively (Godin *et al.*, 2010). The gross energy of each sample was calculated using a coefficient of 17.17 kJ/g for starch and NDF, 23.45 kJ/g for proteins (defined as N $\times$ 6.25) and 38.94 kJ/g for EE (Vaissaire, 2018). ED of DM, starch, nitrogen, EE, ash, NDF, ADF, ADL and energy was calculated for each container using Equation 1, where A stands for the considered macronutrient:

$$\text{Estimated digestibility of } A = \frac{\text{Weight of } A \text{ in distributed feed} - \text{Weight of } A \text{ in frass}}{\text{Weight of } A \text{ in distributed feed}} \times 100 \quad (1)$$

For each container, a sample of 30 larvae was weighed on a precision balance (Sartorius AZ153, Göttingen, Germany, ± 0.001 g) to calculate the mean individual fresh weight gain and estimate the final number of larvae. The survival rate was calculated as the ratio between final and initial numbers of BSFL. Feed conversion ratio (FCR dry to fresh) was calculated as the ratio between DW of feed distributed and fresh larval weight gain. Larvae were killed by freezing at -20°C. Due to improper sample management, no further analysis was performed on Trial 1 larvae. Larvae from Trial 2 were oven-dried at 60°C until constant weight to calculate their DM content. Mean individual dry weight gain and FCR on a dry basis (FCR dry to dry) were also calculated for Trial 2.

2.4. Statistical analysis

Collected data were analysed using R (version 4.1.2., R Core Team, 2021). The effect of areal density and trial were assessed by ANCOVA and two-way ANOVA, with areal density considered respectively as a quantitative or qualitative variable. For larval DM content, larval DW gain and FCR (dry to dry), the effect of areal density was assessed by linear regression and one-way ANOVA. The Bayesian Information Criterion (BIC) was used to select the best model. As the effect of the trial cannot directly be related to a difference in treatment, it was considered a random effect. Further treatments focused only on the impact of areal density and considered all data together. Considering the pattern of scatter plots of ED of DM, starch, nitrogen, EE, ash and energy, and of FCR (dry to fresh), the effect of areal density was modelled for each variable using the nonlinear logistic model ‘logit’ from the ‘drc’ package. The equation of the model is Equation 2, with $-1/b$ the slope for $X=0$, c the lower asymptote, d the upper asymptote and e the inflexion point:

$$Y = c + \frac{d-c}{1+\exp(b(X-e))} \quad (2)$$

The logistic model was used to estimate the value of the asymptote and the areal densities required to reach 95% and 99% of the asymptotic performance (AD95 and AD99, respectively). Logistic models were also proposed for each trial separately.

3. Results

3.1. Estimated digestibility

Figure 1 presents mean values and standard deviations of ED for each trial and areal density. For all macronutrients except NDF, ADF and ADL, two-way ANOVA provided lower BIC for ED than ANCOVA (Table 3). Areal density and interaction between density and trial had a significant effect ($P < 0.05$) on ED of DM, starch, nitrogen, EE, ash and energy. In contrast, trial significantly impacted only the ED of DM, starch, ash and energy. The ANCOVA model resulted in the lowest BIC for ED of NDF and ADL, with a significant effect of trial for both, and of the interaction between trial and areal density for NDF only. Neither the ANCOVA nor the ANOVA model was significantly explanatory for ED of ADF.

In both trials, increasing areal density between 0 and 11.4 larvae/cm² positively affected ED of DM (from $3.0 \pm 2.0\%$ to $79.7 \pm 1.8\%$), starch (from $24.6 \pm 7.8\%$ to $98.7 \pm 1.2\%$), nitrogen (from $-12.3 \pm 3.4\%$ to $78.4 \pm 2.0\%$), EE (from $9.9 \pm 15.3\%$ to $94.8 \pm 1.4\%$), ash (from $-0.6 \pm 1.6\%$ to $60.8 \pm 4.4\%$) and energy (from $11.0 \pm 2.3\%$ to $80.3 \pm 1.2\%$). Higher densities showed similar ED values, resulting in an asymptotic trend, except for ED of ash that slightly decreased between 11.4 and 28.6 larvae/cm² from $60.8 \pm 4.4\%$ down to $54.0 \pm 4.9\%$. Asymptotic ED determined by the logistic model were 80.3%, 99.0%, 78.7%, 95.3%, 58.4% and 80.6% for DM, starch, nitrogen, EE, ash and energy respectively (Table 4).

Areal density had no significant effect on ED of NDF, ADF and ADL. Therefore, no modelisation was proposed for these variables. The mean temperature recorded in the climatic chamber was 27.1°C for Trial 1 and 28.1°C for Trial 2.

Figure 1: (A) Mean estimated digestibility of dry matter, (B) starch, (C) nitrogen, (D) ether extract, (E) ash, (F) energy, (G) NDF (Neutral Detergent Fibre), (H) ADF (Acid Detergent Fibre) and (I) ADL (Acid Detergent Lignin) in black soldier fly larvae fed chicken feed and reared in seven days at varying densities in Trial 1 (n=2) and Trial 2 (n=3). Error bars: standard deviation.

Table 3: P-values of the linear models

Variable	Best model	P-value of the model	Areal density	Trial	Interaction
ED of dry matter	Two-way ANOVA	$< 2.2E-16$ ***	$< 2.2E-16$ ***	$1.56E-04$ ***	$6.18E-03$ **
ED of starch	Two-way ANOVA	$< 2.2E-16$ ***	$< 2.2E-16$ ***	$4.86E-08$ ***	$5.02E-07$ ***
ED of nitrogen	Two-way ANOVA	$< 2.2E-16$ ***	$< 2.2E-16$ ***	$1.28E-01$	$2.92E-02$ *
ED of ether extract	Two-way ANOVA	$< 2.2E-16$ ***	$< 2.2E-16$ ***	$3.59E-01$	$4.13E-03$ **
ED of ash	Two-way ANOVA	$< 2.2E-16$ ***	$< 2.2E-16$ ***	$8.91E-05$ ***	$4.60E-02$ *

ED of energy	Two-way ANOVA	< 2.2E-16 ***	< 2.2E-16 ***	9.43E-05 ***	1.54E-02 *
ED of NDF	ANCOVA	3.04E-03 **	4.95E-01	9.96E-04 ***	8.44E-02
ED of ADF	ANCOVA	1.81E-01	1.17E-01	9.32E-01	8.39E-01
ED of ADL	ANCOVA	7.46E-05 ***	1.47E-01	1.11E-04 ***	5.95E-03 **
Survival	ANCOVA	3.27E-06 ***	1.08E-02 *	7.39E-04 ***	5.46E-05 ***
Larval dry matter content	One-way ANOVA	9.35E-09 ***	9.35E-09 ***	-	-
FCR (dry to fresh)	Two-way ANOVA	< 2.2E-16 ***	< 2.2E-16 ***	1.40E-04 ***	1.60E-02 *
FCR (dry to dry)	One-way ANOVA	3.66E-10 ***	3.66E-10 ***	-	-
Individual fresh weight gain	Two-way ANOVA	< 2.2E-16 ***	< 2.2E-16 ***	1.61E-05 ***	3.35E-04 ***
Individual dry weight gain	One-way ANOVA	1.44E-10 ***	1.44E-10 ***	-	-

Level of signification: $P < 0.001$: ***; $P < 0.01$: **; $P < 0.05$: *. ED: Estimated Digestibility; NDF: Neutral Detergent Fibre; ADF: Acid Detergent Fibre; ADL: Acid Detergent Lignin; FCR: Feed Conversion Ratio.

Table 4: Parameters (intercept and standard error) of the logistic model used to describe the effect of areal density on studied variables, and minimal estimated areal density to reach 95% and 99% of the asymptotic value (AD95 and AD99, respectively).

Variable	P-value of the model	b	c	d	e	AD95 (larvae/cm ²)	AD99 (larvae/cm ²)
ED of dry matter	0.00152	-0.509 (0.07) ***	-14.507 (9.076)	80.252 (1.268) ***	2.839 (0.46) ***	8.97	12.19
ED of starch	0.00117	-0.839 (0.213) ***	23.778 (5.149) ***	98.975 (2.344) ***	4.731 (0.287) ***	7.90	9.88
ED of nitrogen	0.06936	-0.611 (0.081) ***	-49.254 (17.129) **	78.601 (1.109) ***	1.461 (0.521) **	7.11	9.78
ED of ether extract	0.66621	-0.644 (0.115) ***	-14.538 (15.274)	95.259 (1.486) ***	1.93 (0.544) **	6.74	9.29
ED of ash	0.07319	-0.742 (0.099) ***	-7.784 (4.613)	58.386 (0.993) ***	2.801 (0.272) ***	6.95	9.17
ED of energy	0.00009	-0.591 (0.083) ***	0.956 (6.233)	80.576 (1.179) ***	3.171 (0.348) ***	8.13	10.92
FCR (dry to fresh)	0.03065	0.585 (0.064) ***	0.784 (0.019) ***	19.068 (32.653)	-2.012 (3.671)	8.48	11.23

Level of signification: $P < 0.001$: ***; $P < 0.01$: **; $P < 0.05$: *. ED: Estimated Digestibility; FCR: Feed Conversion Ratio.

3.2. Life history traits and conversion efficiency

Figure 2 presents the results of life history traits and conversion efficiency. The ANCOVA model better explained survival rate. In contrast, the ANOVA model was more explanatory for

larval DM content, FCR (dry to fresh), FCR (dry to dry), individual FW gain and individual DW gain. For all these indicators, trial, areal density and the interaction of both had a significant effect, except for larval DM content, FCR (dry to dry) and individual DW gain, which were evaluated only in Trial 2.

Figure 2: (A) Survival rate, (B) larval dry matter content, (C) FCR (feed conversion ratio) on a fresh basis, (D) FCR on a dry basis, (E) individual fresh weight gain and (F) individual dry weight gain of black soldier fly larvae fed chicken feed and reared in seven days at varying densities in Trial 1 (n=2) and Trial 2 (n=3). Error bars: standard deviation.

The mean survival rate ranged between $100.2 \pm 2.5\%$ and $85.4 \pm 3.8\%$ and declined with increasing areal density. Values above 100% are a computational artefact due to a lack of accuracy in estimating the initial and final number of BSFL. FCR (dry to fresh) was continuously improved by increasing areal density, while FCR (dry to dry) only decreased between 2.29 and 11.43 larvae/cm² and worsened at higher densities. FCR (dry to fresh) was modelled with a logistic regression: lower asymptote, AD95 and AD99 were 0.78, 8.48 larvae/cm² and 11.23 larvae/cm², respectively. No converging logistic equation could be modelled for FCR (dry to dry). Larval DM content, individual FW gain and individual DW gain reached their maximal value at 4.00 larvae/cm² and decreased with rising areal density.

4. Discussion

This study proposed a method based on frass analysis and variation of larval density to assess maximal ED in BSFL. The results showed that the ED of DM, starch, nitrogen, EE, ash and energy is positively affected by areal density, following a sigmoidal pattern that can be modelled with a logistic curve. High areal densities allowed more BSFL to collectively feed on the same amount of CF, leading to higher ED values, up to an asymptote that indicates the maximal fraction of feed BSFL can potentially digest. Indigestible macronutrients remaining in the frass were not biologically available for BSFL either because they are physically embedded in a resistant structure (such as proteins in a vegetal cell wall) or because larvae do not have the appropriate enzymatic equipment. In CF, BSFL were able to digest up to 80.3%, 99.0%, 78.6%, 95.3%, 58.4% and 80.6% of DM, starch, nitrogen, EE, ash and energy respectively. ED of DM in CF in the literature ranged between 49.9 and 84.8% (Table 5). Brits explored the effect of density on ED of DM in kitchen waste and, although the experimental set-up did not allow for a clear asymptotic trend, a maximal value of 35.5% was obtained. This result suggests CF to be more digestible than kitchen waste. Values of ED of starch, nitrogen, EE, ash and energy in CF are unavailable in the literature, but some have been reported in other types of feed. ED of nitrogen ranged from 27% (in pig manure) to 72.4% (in soybean curd residue). In a mixture of yeast, wheat and potato by-products, ED of starch, nitrogen, lipids and energy were 55.8%, 37.9%, 72.9% and 46.8%, respectively (Table 5). Values of ED from the present study were the highest reported for these macronutrients. However, these results should be compared with caution due to a lack of standardisation in experimental conditions between studies, namely in terms of larval density and feeding time (Bosch *et al.*, 2020).

332 Table 5: Comparison of estimated digestibility in black soldier fly larvae (BSFL) for different types of feed and rearing conditions.

333 FW: fresh weight; DW: dry weight; DM: dry matter; CM: cattle manure; SBR: soybean curd residue.

Type of feed	Number of BSFL	FW of diet (g)	Moisture content	Larval density			Feeding time (day)	Number of feed supplies	Estimated Digestibility											Reference
				Areal density (larvae/cm ²)	Mass density (larvae/kg DW)	Feed availability (mg DW/larva)			DM	Carbon	Nitrogen	Starch	Lipids	Energy	Phosphorus	Potassium	Cellulose	Hemicellulose	Lignin	
Chicken feed	600	1,300	61.5%	-	1,199	834	15	1	70.0%	-	-	-	-	-	-	-	-	-	-	(Gobbi <i>et al.</i> , 2013)
	200	310.7	58.8%	0.56	1,563	640	-	Several	84.8%	-	-	-	-	-	-	-	-	-	-	(Lalander <i>et al.</i> , 2019)
	80	48.0	60.0%	1.81	4,167	240	9	3	67.7%	-	-	-	-	-	-	-	-	-	-	(Gold <i>et al.</i> , 2020b)
	500	400	70.0%	2.40	4,167	240	9	4	49.9%	-	-	-	-	-	-	-	-	-	-	(Broeckx <i>et al.</i> , 2021)
	80	41.0	60.0%	2.80	4,878	205	9	1	54.7%	-	-	-	-	-	-	-	-	-	-	(Cohn <i>et al.</i> , 2022)
	80	32.8	60.0%	2.80	6,107	164	9	1	58.5%	-	-	-	-	-	-	-	-	-	-	
Kitchen waste	330	990	75.1%	3.35	1,340	746	15	1	25.8%	-	-	-	-	-	-	-	-	-	-	(Brits, 2017)
	440	990	75.1%	4.47	1,786	560	15	1	32.9%	-	-	-	-	-	-	-	-	-	-	
	528	990	75.1%	5.36	2,144	467	15	1	35.5%	-	-	-	-	-	-	-	-	-	-	
	660	990	75.1%	6.70	2,680	373	15	1	35.0%	-	-	-	-	-	-	-	-	-	-	
	880	990	75.1%	8.93	3,573	280	15	1	33.8%	-	-	-	-	-	-	-	-	-	-	
	1,320	990	75.1%	13.40	5,359	187	15	1	34.2%	-	-	-	-	-	-	-	-	-	-	
CM	1,000	1,000	78.4%	-	4,630	216	24.3	1	25.8%	52.0%	32.0%	-	-	-	38.7%	-	51.0%	51.7%	35.0%	(Rehman <i>et al.</i> , 2017)
80% CM + 20% SBR	1,000	1,000	79.2%	-	4,808	208	22.5	1	32.4%	60.0%	34.7%	-	-	-	41.3%	-	51.3%	43.3%	25.0%	

60% CM + 40% SBR	1,000	1,000	79.7%	-	4,926	203	20.6	1	46.5%	62.7%	48.0%	-	-	-	48.0%	-	55.0%	51.7%	32.3%	
40% CM + 60% SBR	1,000	1,000	80.7%	-	5,181	193	21.1	1	56.6%	65.3%	60.0%	-	-	-	50.7%	-	64.7%	63.3%	36.7%	
20% CM + 80% SBR	1,000	1,000	81.5%	-	5,405	185	21.1	1	68.3%	72.0%	72.0%	-	-	-	58.7%	-	69.7%	71.3%	33.3%	
SBR	1,000	1,000	82.3%	-	5,650	177	19.4	1	72.4%	70.7%	73.3%	-	-	-	68.0%	-	68.3%	81.7%	39.3%	
40% CM + 60% SBR	1,000	1,000	76.1%	-	4,179	239	20.6	1	42.0%	-	54.0%	-	47.7%	-	-	-	49.6%	55.1%	24.0%	(Rehman <i>et al.</i> , 2019)
Pig manure	22,500	18,830	76.1%	4.50	5,000	200	9	1	22.0%	27.0%	47.0%	-	-	30.0%	9.0%	6.0%	-	-	-	(Parodi <i>et al.</i> , 2021)
47% yeast concentrate + 47% wheat and potato by-product + 6% binding agent	10,000	4,000	71.7%	6.67	8,834	113	7	1	42.0%	45.8%	37.9%	55.8%	72.9%	46.8%	27.5%	12.6%	-	-	-	Adapted from Parodi <i>et al.</i> , 2020

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Asymptotic ED in BSFL can also be compared to Apparent Digestibility and Approximate Digestibility in other insect species. Both vary significantly with feed, insect species, rearing conditions and larval instar. In mealworm larvae (*Tenebrio molitor*), wheat bran has an Approximate Digestibility of DM of 46% and an Apparent Digestibility of nitrogen, starch and lipids of 54%, 59% and 73%, respectively (Waldbauer, 1968). The Apparent Digestibility of wheat in growing pig and ruminant is 90% and 88% for organic matter, 84% and 70% for nitrogen, 24% and 63% for lipids, and 88% and 86% for energy, respectively (Sauvant *et al.*, 2002). Hence BSFL show strong potential to digest high-quality feed such as CF, comparable to what can be observed in conventional livestock species.

In contrast, ED of NDF, ADF and ADL were not affected by areal density. Mean ED values ($22.6 \pm 5.9\%$, $6.6 \pm 6.1\%$ and $11.8 \pm 15.6\%$, respectively) were below those of other macronutrients, suggesting that BSFL are inefficient at digesting fibres. These results are consistent with reported digestive activity in BSFL, which are high for amylase, lipase, and protease but low for cellulase (Intayung *et al.*, 2021; Sun *et al.*, 2021). This also concurs with BSFL developing well on substrates rich in starch and proteins and having low growth performance on fibre-rich substrates (Gold *et al.*, 2018). However, these very low ED of fibres are inconsistent with results obtained on soybean curd residue, cattle manure and mixtures of both, where ED varied between 49.6% and 69.7% for cellulose, 43.3% and 81.7% for hemicellulose, 24.0% and 39.3% for lignin (Rehman *et al.*, 2017). This difference could indicate that fibre fractions are less digestible in CF than in cattle manure and soybean curd residue. However, low fibre ED from the present study could also result from an inappropriate dosage method. Indeed, it has been reported that the Van Soest method was not specific to vegetal fibre fractions and that ADF could be used to estimate chitin content in shellfish (Stelmock *et al.*, 1985). Although insect and shellfish chitin have structural differences (Brigode *et al.*, 2020), it can be assumed that at least part of the chitin from dead BSFL and exuviae in frass was quantified by the Van Soest method along with plant fibres. It remains unclear whether the Van Soest method is appropriate for estimating residual plant fibre content in frass. Other dosage methods should be explored, such as a pre-treatment of frass samples with chitinase. As chitin contains nitrogen, measuring frass chitin content would also allow a more accurate nitrogen ED determination.

Although differences were detected between the results of both trials, presumably caused by a slightly higher temperature in Trial 2 (28.1°C) compared to Trial 1 (27.1°C), the overall pattern of the data and the values of asymptotic ED remained the same, except for ash. This supports the opinion that asymptotic ED is a robust indicator of the ability of BSFL to digest their feed, little to not impacted by temperature.

With 0 larvae/cm², ED was positive for DM ($3.0 \pm 2.0\%$), starch ($24.6 \pm 7.8\%$), EE ($9.9 \pm 15.3\%$) and energy ($11.0 \pm 2.3\%$). This is likely due to the activity of microorganisms in the substrate that partly digested it to produce microbial biomass. BSFL activity modifies the microbial community and both affect each other in a process called “co-conversion” (Cifuentes *et al.*, 2020; Xiao *et al.*, 2018), with BSFL consuming and digesting microbial biomass (Gold *et al.*, 2018). Conversely, 0 larvae/cm² density resulted in negative ED for nitrogen ($-12.3 \pm 3.4\%$) and ash ($-0.7 \pm 1.6\%$), which indicated a negative balance between macronutrients removed and added to the substrate. Negative ED of nitrogen could result from diazotrophs in the feed that transformed atmospheric dinitrogen into microbial proteins or ammonia released in the substrate (Chan *et al.*, 1994). Without BSFL, the substrate kept a high moisture content: therefore, it can also be hypothesised that it absorbed the gaseous and hydrophilic ammonia produced by BSFL of other containers in the same climatic chamber. The increase in ash

concentration at null density is very likely due to the mineral content of tap water used for feed preparation. ED of ash reached its maximum at 11.4 larvae/cm² but decreased with higher densities, indicating that the substrate accumulated minerals. This might be an experimental artefact. As starter larvae were not completely free from frass, high-density treatments were also inoculated with more frass than those of lower areal density, hence receiving more undigested ash from pre-experimental feeding. Higher frass ash content at high larval density could also result from a higher number of exuviae shed by BSFL, which bioaccumulate minerals in the cuticle in the form of calcium carbonate (Rubio *et al.*, 2021).

Life history traits values obtained were consistent with data from the literature on chicken feed: (Liu *et al.*, 2019) reported survival rates between 40 and 93% and (Gold *et al.*, 2018) indicated that prepupal FW could range between 99 and 184 mg. Larval DM content varied in the range of 17.9% to 38.8% reported by Bosch *et al.* In the present study, high areal density caused stronger competition for feed resources, hence lower survival rate and individual FW and DW gain, as already observed (Brits, 2017; Jones *et al.*, 2019; Opare *et al.*, 2022). Dead BSFL could not be visually detected after seven days of feeding and were presumably consumed by other larvae or decomposed into the substrate. Similar adverse effects of density on survival or growth have been reported in other insect species such as *Drosophila melanogaster* (Henry *et al.*, 2020), *Musca domestica* (Kökdener and Kiper, 2020) or *Tenebrio molitor* (Deruytter and Coudron, 2021). Impaired larval DM content observed in the present work at high density suggests that limited access to feed caused BSFL to use their body energy resources by oxidising their fat reserve (Opare *et al.*, 2022). However, FCR (dry to fresh) was improved by increasing density, indicating that BSFL performance as a group outweighed each individual underperformance. The same occurred with FCR (dry to dry) but only up to 11.4 larvae/cm². Interestingly, low densities ranging from 0.31 to 2.47 larvae/cm² are reported not to affect individual FW (Barragan-Fonseca *et al.*, 2018), presumably because the density was too low to induce competition. These contrasting findings suggest that, although increasing density results in competition and possible accumulation of toxic metabolic waste in the substrate, it also supports larval growth via mechanisms such as increased temperature and aeration of the substrate due to BSFL activity, cooperation through extra-oral digestion, or facilitated development of beneficial microbiota (Barragan-Fonseca *et al.*, 2018; Parra Paz *et al.*, 2015).

Differences between AD95 of ED of EE, ash, nitrogen, starch, energy and DM (6.74, 6.95, 7.11, 7.90, 8.13 and 8.97 larvae/cm², respectively) indicate that calculated optimal density to fully exploit the nutritional potential of a diet depends on the macronutrient considered. These differences could be related to the amount of each macronutrient in CF: it contained far less EE than starch and it might explain why EE requires fewer larvae to be optimally digested. Starch was the main component and the primary source of energy in CF. Therefore, AD95 of starch, DM and energy are relatively close. AD95 of FCR (dry to fresh) (8.48 larvae/cm²) is also close, which highlights that starch is an essential driver of BSFL weight gain (Barragan-Fonseca *et al.*, 2021).

Optimal areal density should be chosen based on the aim of the BSFL rearing facility (Myers *et al.*, 2008). FCR (dry to dry) should be the main driver for manufacturers producing insect proteins. According to present data, densities around 11.43 larvae/cm² would produce the best output. On the other hand, waste management stakeholders use BSFL to reduce the weight of treated substrate and should be driven by the ED of DM. Therefore, areal density around AD95 of ED of DM (8.97 larvae/cm²) is to be used. However, optimal densities should also be based on economic considerations as starter BSFL can be costly to produce. Insect producers should

seek a trade-off between maximising the animal performance and minimising the cost of operations.

The results of this study were obtained on CF with a 68% moisture content and in specific containers. Further investigations should focus on evaluating asymptotic ED and AD95 for different types of feed, but also for different feeding conditions likely to affect BSFL digestion, such as moisture content (Dzepe *et al.*, 2020), temperature (Harnden and Tomberlin, 2016), shape and size of the container (Miranda *et al.*, 2021), particle size (Brits, 2017) and addition of beneficial microorganisms (Zhang *et al.*, 2021). The latter represents an excellent opportunity to improve fibre digestion by using cellulolytic bacteria or fungi (Peguero *et al.*, 2022). ED of macronutrients of a standard diet could also be an insightful indicator when developing a genetic selection strategy for *H. illucens* (Eriksson and Picard, 2021), as Zhou *et al.* showed that strain had an impact on ED of DM.

High-density containers reached a plateau of ED corresponding to the maximal amount of biologically available macronutrients. However, it remains unclear whether the remaining macronutrients were not digested because they were undigestible or because the substrate conditions evolved and inhibited further ingestion. Indeed, at the end of the seven-day feeding period, frass in high-density containers was dry and powdery. This texture is inconsistent with BSFL mouthparts, adapted to ingest semi-liquid feed (Bruno *et al.*, 2020). Further investigations on the effect of water supplementation during growth on asymptotic ED are needed. Similarly, as frass was moist and sticky in low-density containers, BSFL might have been able to digest it further with an extended feeding period, leading to an ED closer to its maximal value. This hypothesis is supported by the ED of DM of 67.7% reported on BSFL fed CF with a density of 1.81 larvae/cm² during nine days (Gold *et al.*, 2020b), far above the value of 29.7% in seven days predicted by the model of the present study.

ED calculation is based on chemical analysis of distributed feed and frass, which is a mixture of non-ingested feed, faeces, urines, dead individuals and exuviae. This indicator is suitable for insects living in their feed like BSFL. Yet a more accurate digestibility indicator could be developed based on what is done in other insect or livestock species. Using an indigestible marker (Barret *et al.*, 2005) and collecting pure excreta would allow for calculating ingesta and Approximate Digestibility (Equation 3). This could be done with an excreta collection device (Gold *et al.*, 2020a), by separating the top and bottom ends of BSFL by a perforated wall to allow them to feed from a compartment and release excreta in another one, or by putting clean BSFL in an empty container after a feeding period. However, all options lead to tiny excreta samples, are impractical on many larvae, and possibly stressful with an impact on digestion. Dissection of BSFL digestive tract to collect faeces should be avoided as the collected material may not be fully digested (Guillaume *et al.*, 1999). In insects producing excreta that can be separated from other frass constituents, uric acid can be dosed in excreta to calculate Apparent Digestibility (Equation 4), similarly to what is done in birds (Waldbauer, 1968). Yet the molecular form of nitrogen in metabolic waste excreted by Diptera remains unclear (Henry *et al.*, 2020) and further work is needed on urinary wastes in frass and their degradation products, i.e. uric acid, allantoin, urea, ammonia, nitrite and nitrate (Barragan-Fonseca *et al.*, 2021). The material produced in the digestive tract, also called endogenous faeces (microbial biomass, enzymes and gut cells), could also be quantified to calculate True Digestibility (Equation 5) (Barret *et al.*, 2005). However, this indicator was developed on animals producing an important amount of pure easy-to-collect faeces, such as poultry or pigs and is complex to assess in small insects living in groups inside their feed (Figure 3). *In vitro* simulation of BSFL digestion is

limited by understanding the digestive tract's enzymatic and microbial activity (Gold *et al.*, 2020c).

$$\text{Approximate digestibility} = 1 - \frac{\text{Weight of excreta}}{\text{Weight of ingesta}} \quad (3)$$

$$\text{Apparent digestibility} = 1 - \frac{\text{Weight of excreta} - \text{Weight of urines}}{\text{Weight of ingesta}} \quad (4)$$

$$\text{Apparent digestibility} = 1 - \frac{\text{Weight of excreta} - \text{Weight of urines} - \text{Weight of endogenous faeces}}{\text{Weight of ingesta}} \quad (5)$$

Figure 3: Breakdown of feed distributed to black soldier fly larvae during digestion

5. Conclusion

The present work proposes a method to evaluate BSFL feed efficiency via asymptotic ED, a new indicator defined as the maximal fraction of initial macronutrients that are not in the frass at the end of the feeding period. ED of major macronutrients are assessed on a standard diet with a focus on the effect of larval areal density. Asymptotic ED corresponds to the maximal proportion of each macronutrient that BSFL can potentially digest. Therefore, when formulating a diet for BSFL based on several ingredients, crude macronutrient content multiplied by the asymptotic ED of that macronutrient should be considered for each ingredient, rather than the sole crude macronutrient content. Further studies should focus on determining asymptotic ED of macronutrients in different types of ingredients that could be used as BSFL feed (such as agricultural side-streams, food waste or animal manure) and evaluate the effect of rearing conditions on asymptotic ED.

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